Containerization-and-DevOps

🚀 Class Task

🛠️ Tools Used

- Kubernetes (kubectl)
- k3d
- Docker

⚙️ Steps

Create Apache Pod

```
kubectl run apache-pod --image=httpd
```

Check Pod Status

```
kubectl get pods
```

Describe Pod

```
kubectl describe pod apache-pod
```

Port Forward Pod

```
kubectl port-forward pod/apache-pod 8081:80
```

Delete Pod

```
kubectl delete pod apache-pod
```

Create Apache Deployment

```
kubectl create deployment apache --image=httpd
```

Check Deployment & Pods

```
kubectl get deployments
kubectl get pods
```

Expose Deployment

```
kubectl expose deployment apache --port=80 --type=NodePort
```

Port Forward Service

```
kubectl port-forward service/apache 8082:80
```

Scale Deployment

```
kubectl scale deployment apache --replicas=2
```

Describe Pod (Replica)

```
kubectl describe pod <pod-name>
```

Update Image

```
kubectl set image deployment/apache httpd=httpd
```

Execute Inside Pod

```
kubectl exec -it <pod-name> -- /bin/bash
```

Delete Pod (Auto Recreation)

```
kubectl delete pod <pod-name>
kubectl get pods -w
```

Delete Deployment & Service

```
kubectl delete deployment apache
kubectl delete service apache
```

⚠️ Notes

- Pod initially in `ContainerCreating` / `Pending` state before running
- Port forwarding fails if pod is not yet running
- Service creation duplicated → "already exists" error handled
- Deployment ensures pod auto-recreation after deletion 🔁
- `<pod-name>` must be replaced with actual pod name (common mistake 👀)

✅ Outcome

- Apache pod and deployment successfully created
- Scaling tested with multiple replicas
- Service exposed and accessed via port forwarding
- Pod lifecycle (create → delete → recreate) verified

- Deployment cleanup completed successfully

📸 Screenshots

```
sristee@Sristee:/mnt/c/Users/srist$ kubectl run apache-pod --image=httpd
kubectl get pods
kubectl describe pod apache-pod
pod/apache-pod created
NAME                    READY   STATUS              RESTARTS   AGE
apache-pod              0/1     ContainerCreating   0          1s
web-54466ccbf7-49s64    0/1     Unknown             0          6d
web-54466ccbf7-8f584    0/1     Unknown             0          6d
web-54466ccbf7-jzpvf    0/1     Unknown             0          6d1h
Name:             apache-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             k3d-mycluster-server-0/172.20.0.2
Start Time:       Wed, 25 Mar 2026 06:04:06 +0000
Labels:           run=apache-pod
Annotations:      <none>
Status:           Pending
IP:
IPs:              <none>
Containers:
  apache-pod:
    Container ID:
    Image:          httpd
    Image ID:
    Port:           <none>
    Host Port:      <none>
    State:          Waiting
      Reason:       ContainerCreating
    Ready:          False
    Restart Count:  0
    Environment:    <none>
    Mounts:
```

```
sristee@Sristee:/mnt/c/Users/srist$ kubectl port-forward pod/apache-pod 8081:80
Forwarding from 127.0.0.1:8081 -> 80
Forwarding from [::1]:8081 -> 80
^Csristee@Sristee:/mnt/c/Users/srist$ kubectl delete pod apache-pod
pod "apache-pod" deleted from default namespace
sristee@Sristee:/mnt/c/Users/srist$ kubectl create deployment apache --image=httpd
kubectl get deployments
kubectl get pods
deployment.apps/apache created
NAME     READY   UP-TO-DATE   AVAILABLE   AGE
apache   0/1     1            0           0s
web      3/3     3            3           6d1h
NAME                      READY   STATUS              RESTARTS       AGE
apache-7d8d5c5d5-25qkn    0/1     ContainerCreating   0              0s
web-54466ccbf7-49s64      1/1     Running             1 (7m7s ago)   6d
web-54466ccbf7-8f584      1/1     Running             1 (7m7s ago)   6d
web-54466ccbf7-jzpvf      1/1     Running             1 (7m7s ago)   6d1h
sristee@Sristee:/mnt/c/Users/srist$ kubectl expose deployment apache --port=80 --type=NodePort
kubectl port-forward service/apache 8082:80
service/apache exposed
error: unable to forward port because pod is not running. Current status=Pending
sristee@Sristee:/mnt/c/Users/srist$ kubectl expose deployment apache --port=80 --type=NodePort
Error from server (AlreadyExists): services "apache" already exists
sristee@Sristee:/mnt/c/Users/srist$ kubectl get pods
NAME                      READY   STATUS    RESTARTS        AGE
apache-7d8d5c5d5-25qkn    1/1     Running   0               80s
web-54466ccbf7-49s64      1/1     Running   1 (8m27s ago)   6d
web-54466ccbf7-8f584      1/1     Running   1 (8m27s ago)   6d
web-54466ccbf7-jzpvf      1/1     Running   1 (8m27s ago)   6d1h
sristee@Sristee:/mnt/c/Users/srist$ kubectl expose deployment apache --port=80 --type=NodePort
Error from server (AlreadyExists): services "apache" already exists
sristee@Sristee:/mnt/c/Users/srist$ kubectl port-forward service/apache 8082:80
Forwarding from 127.0.0.1:8082 -> 80
```

```
Forwarding from [::1]:8082 -> 80
sristee@Sristee:/mnt/c/Users/srist$ kubectl scale deployment apache --replicas=2=2
kubectl get pods
deployment.apps/apache scaled
NAME                      READY   STATUS              RESTARTS      AGE
apache-7d8d5c5d5-25qkn    1/1     Running             0             8m9s
apache-7d8d5c5d5-7gk5k    0/1     ContainerCreating   0             0s
web-54466ccbf7-49s64      1/1     Running             1 (15m ago)   6d
web-54466ccbf7-8f584      1/1     Running             1 (15m ago)   6d
web-54466ccbf7-jzpvf      1/1     Running             1 (15m ago)   6d1h
sristee@Sristee:/mnt/c/Users/srist$ kubectl describe pod <pod-name>
-bash: syntax error near unexpected token `newline'
sristee@Sristee:/mnt/c/Users/srist$ kubectl describe pod apache-7d8d5c5d5-25qkn
Name:             apache-7d8d5c5d5-25qkn
Namespace:        default
Priority:         0
Service Account:  default
Node:             k3d-mycluster-server-0/172.20.0.2
Start Time:       Wed, 25 Mar 2026 06:10:54 +0000
Labels:           app=apache
                  pod-template-hash=7d8d5c5d5
Annotations:      <none>
Status:           Running
IP:               10.42.0.22
IPs:
  IP:           10.42.0.22
Controlled By:  ReplicaSet/apache-7d8d5c5d5
Containers:
  httpd:
    Container ID:   containerd://701aa9bbce41970b6e48603086f0b52a52af6fe463ebbc47985ab9584fbc5428
    Image:          httpd
    Image ID:       docker.io/library/httpd@sha256:331548c5249bdeced0f048bc2fb8c6b6427d2ec6508bed9c1fec6c57d0b27a60
    Port:           <none>
```

```
  Normal   Started    8m46s                  kubelet            spec.containers{httpd}: Started container httpd
sristee@Sristee:/mnt/c/Users/srist$ kubectl set image deployment/apache httpd=httpd
sristee@Sristee:/mnt/c/Users/srist$ kubectl get pods
kubectl exec -it <pod-name> -- /bin/bash
NAME                      READY   STATUS    RESTARTS      AGE
apache-7d8d5c5d5-25qkn    1/1     Running   0             9m40s
apache-7d8d5c5d5-7gk5k    1/1     Running   0             91s
web-54466ccbf7-49s64      1/1     Running   1 (16m ago)   6d
web-54466ccbf7-8f584      1/1     Running   1 (16m ago)   6d
web-54466ccbf7-jzpvf      1/1     Running   1 (16m ago)   6d1h
-bash: pod-name: No such file or directory
sristee@Sristee:/mnt/c/Users/srist$ ls /usr/local/apache2/htdocs
ls: cannot access '/usr/local/apache2/htdocs': No such file or directory
sristee@Sristee:/mnt/c/Users/srist$ kubectl exec -it apache-7d8d5c5d5-25qkn -- /bin/bash
root@apache-7d8d5c5d5-25qkn:/usr/local/apache2# ls /usr/local/apache2/htdocs
index.html
root@apache-7d8d5c5d5-25qkn:/usr/local/apache2# exit
exit
sristee@Sristee:/mnt/c/Users/srist$ kubectl delete pod apache-7d8d5c5d5-25qkn
kubectl get pods -w
pod "apache-7d8d5c5d5-25qkn" deleted from default namespace
NAME                      READY   STATUS              RESTARTS      AGE
apache-7d8d5c5d5-5d2cg    0/1     ContainerCreating   0             3s
apache-7d8d5c5d5-7gk5k    1/1     Running             0             4m12s
web-54466ccbf7-49s64      1/1     Running             1 (19m ago)   6d
web-54466ccbf7-8f584      1/1     Running             1 (19m ago)   6d
web-54466ccbf7-jzpvf      1/1     Running             1 (19m ago)   6d1h
apache-7d8d5c5d5-5d2cg    1/1     Running             0             6s
sristee@Sristee:/mnt/c/Users/srist$ kubectl delete deployment apachehe
kubectl delete service apache
deployment.apps "apache" deleted from default namespace
service "apache" deleted from default namespace
sristee@Sristee:/mnt/c/Users/srist$
```